

NSDI
April 10, 2018

Andromeda

Performance, Isolation, and Velocity at Scale in Cloud Network Virtualization

Google Cloud

Andromeda Goals

Performance and Isolation

High throughput and low latency, regardless of the actions of other tenants

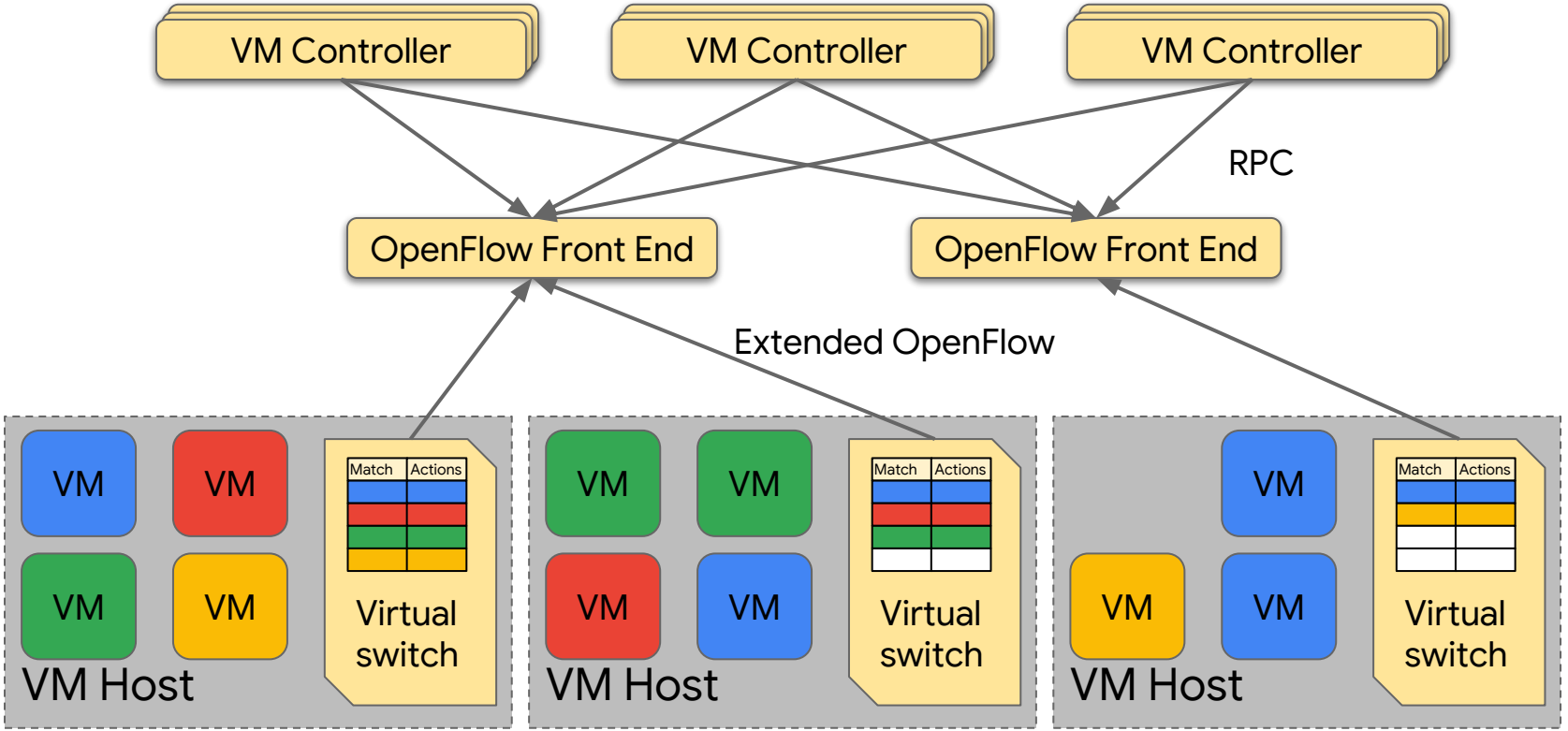
Velocity

Quickly develop and deploy new features and performance improvements

Scalability

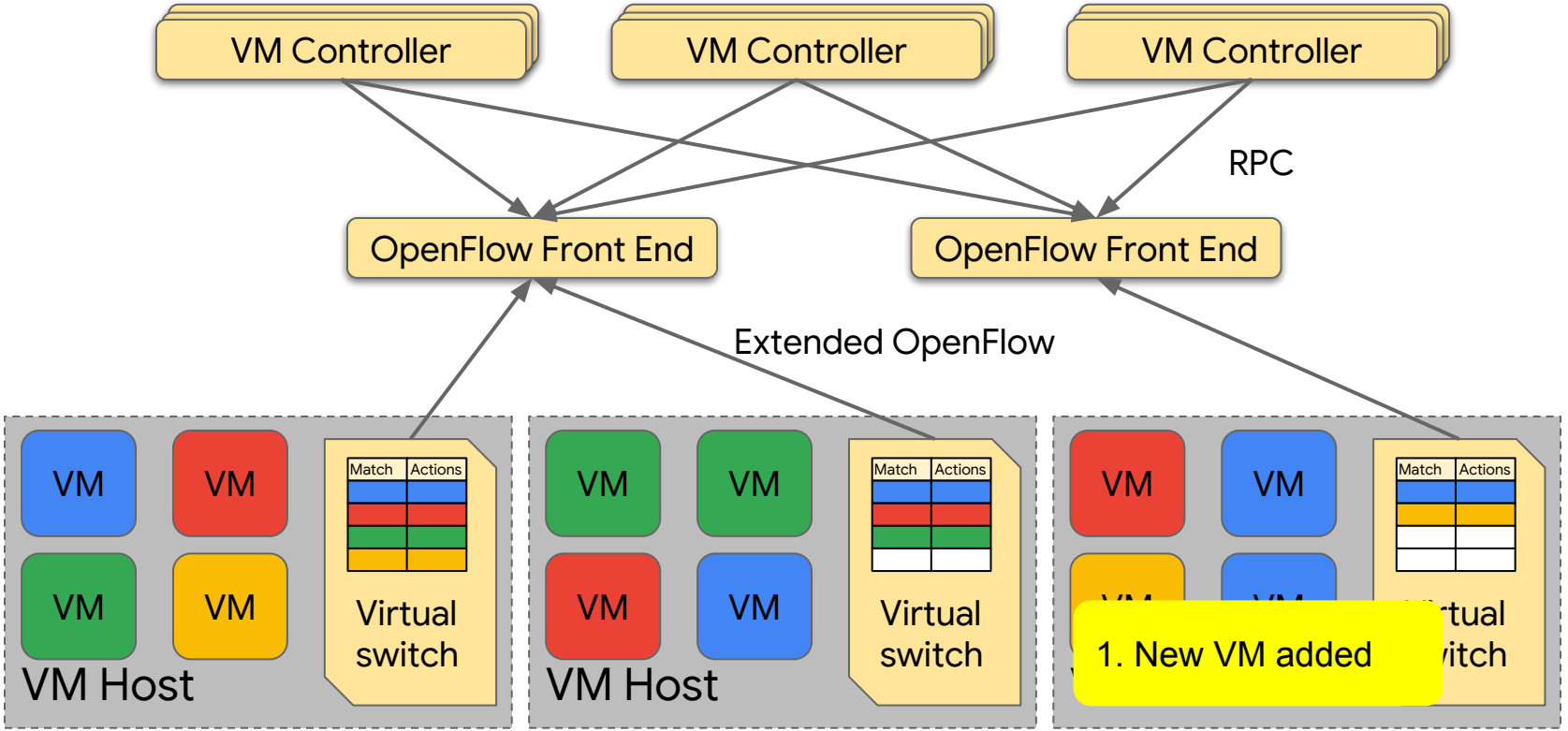
Large networks, many tenants, rapid provisioning

Andromeda Architecture



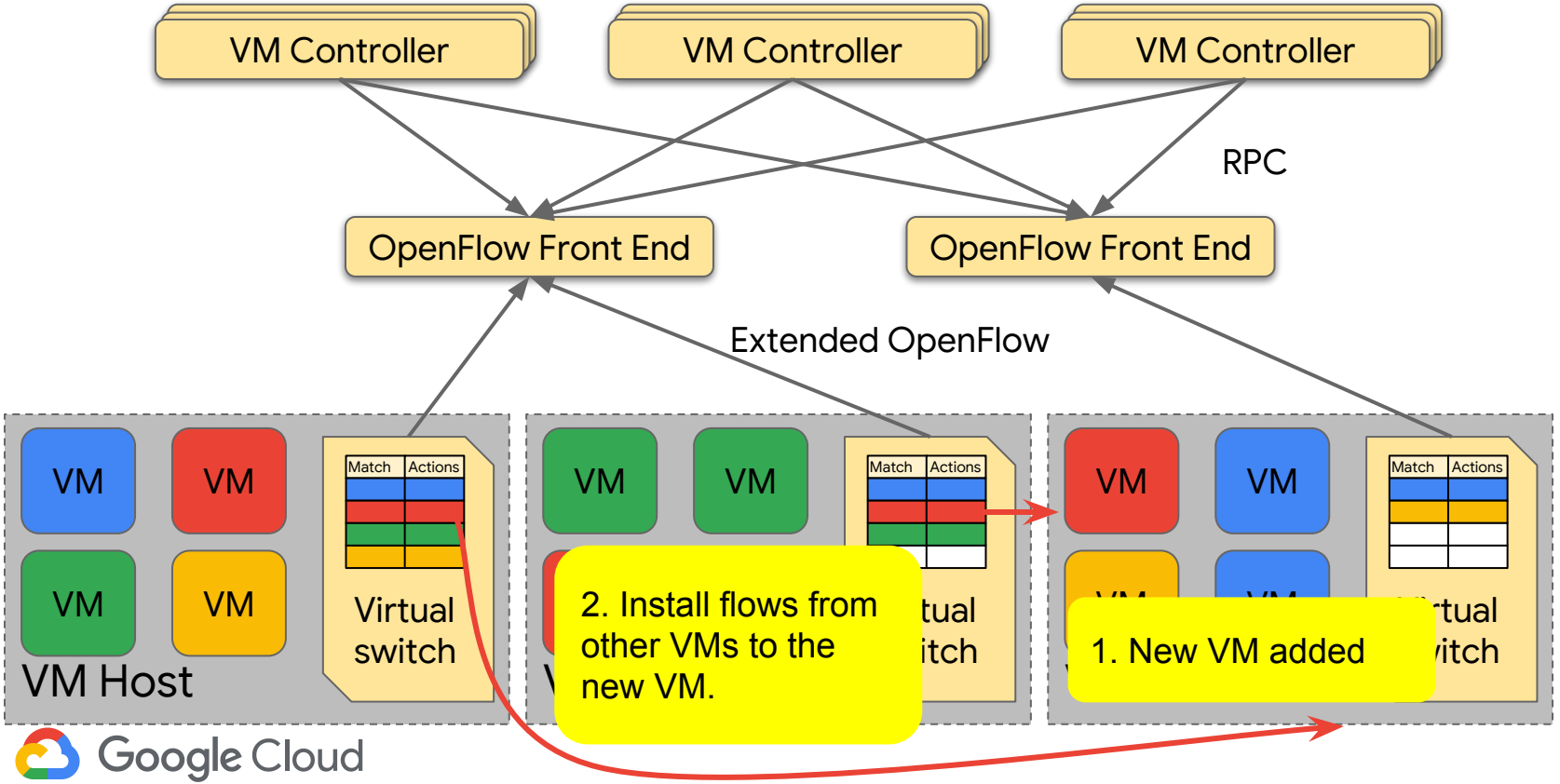
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Andromeda Architecture



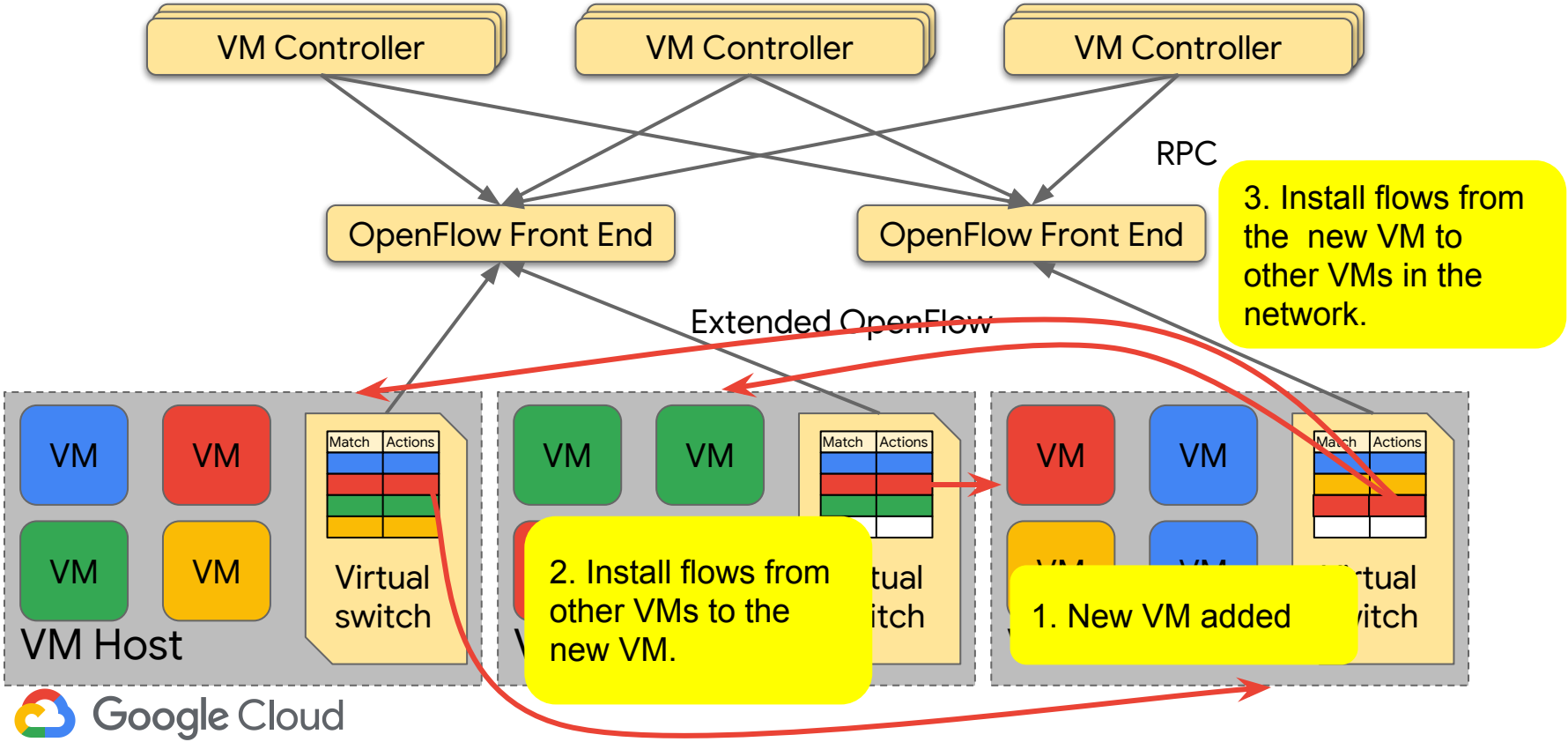
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Andromeda Architecture



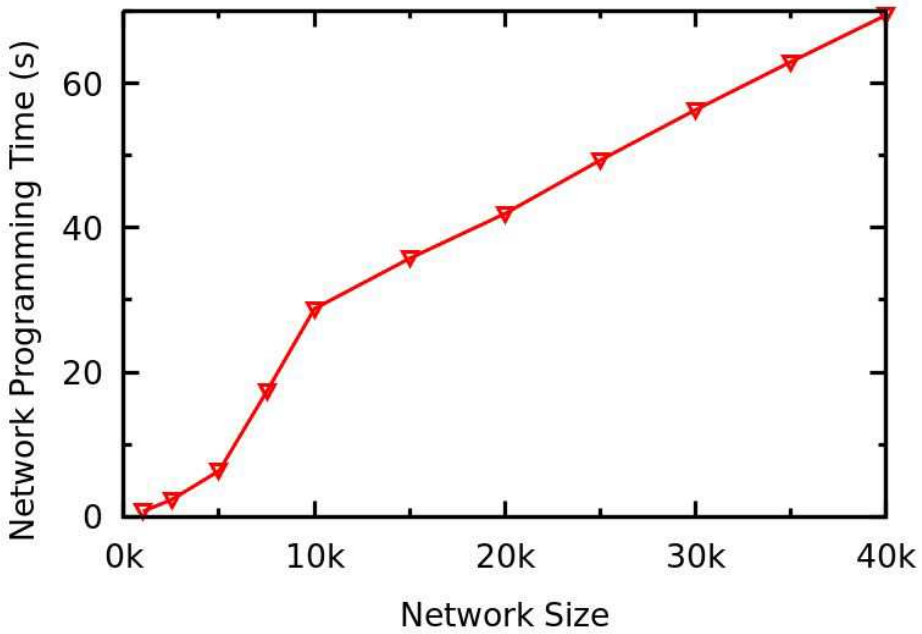
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Andromeda Architecture



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Programming Time for Large Networks



Setup:

- VMs are placed on 10,000 hosts
- 30 VM Controller partitions

Programming time is $O(n \times H)$

n = number of VMs

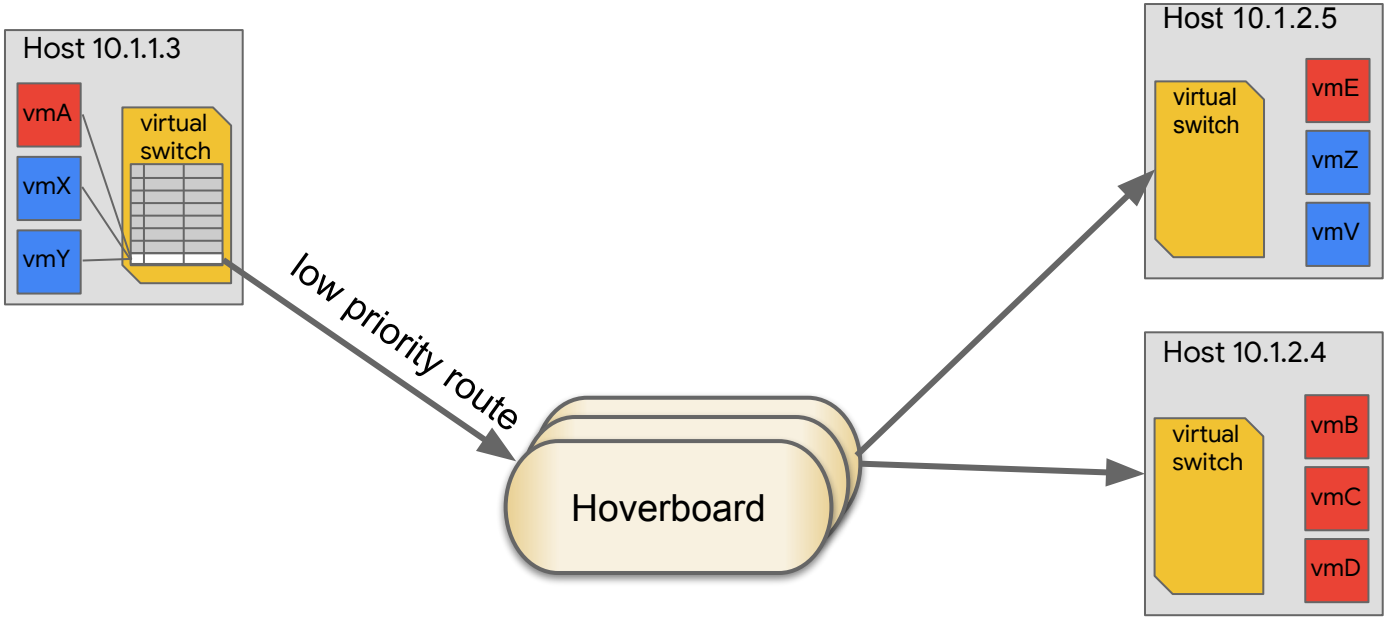
H = number of hosts

Quadratic scaling leads to provisioning challenges

- Control plane CPU and memory
- Dataplane memory

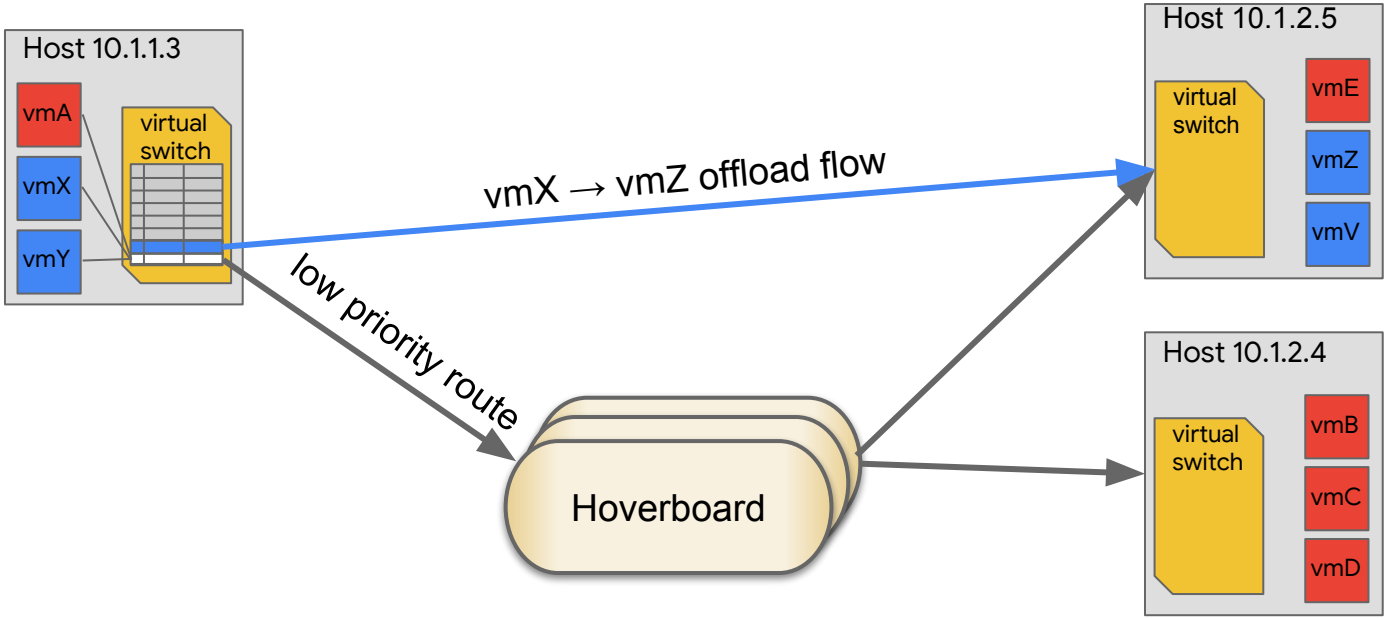
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Scaling with Hoverboards



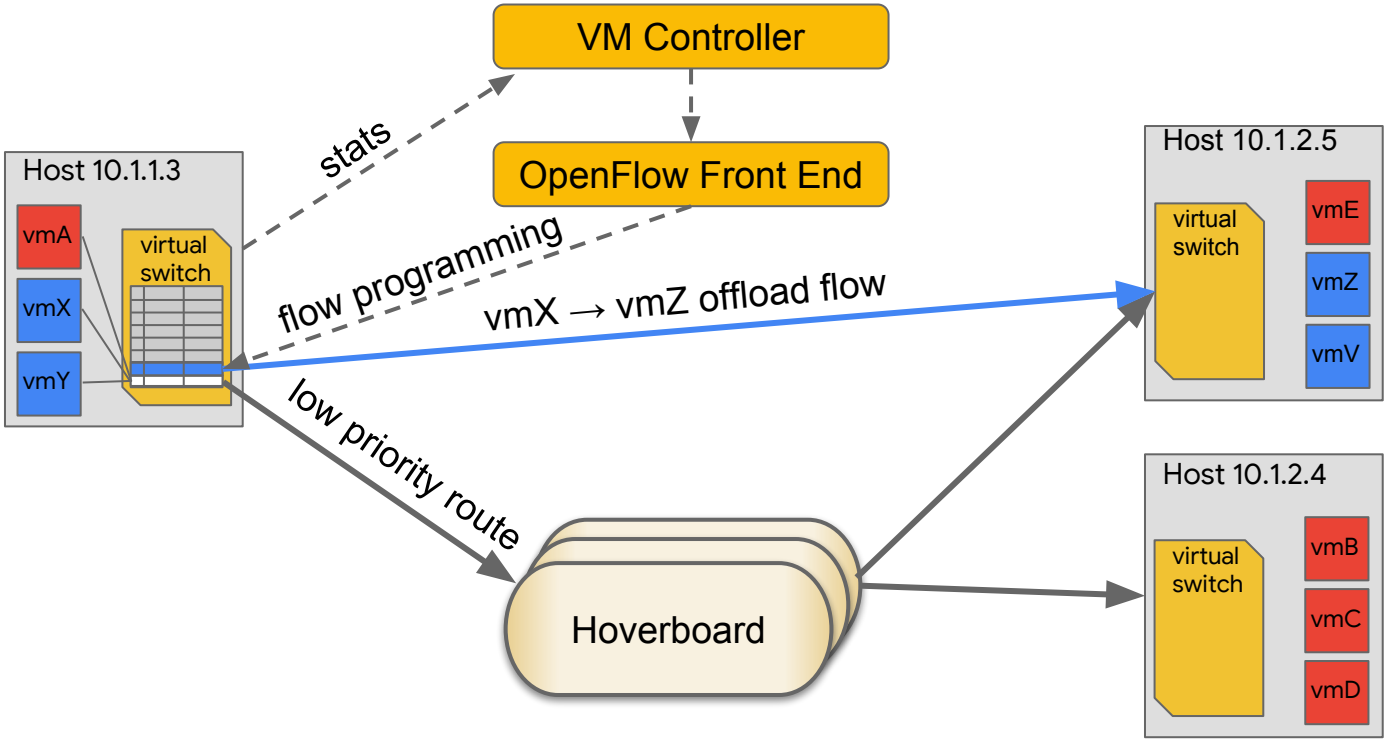
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Hoverboard Offloading



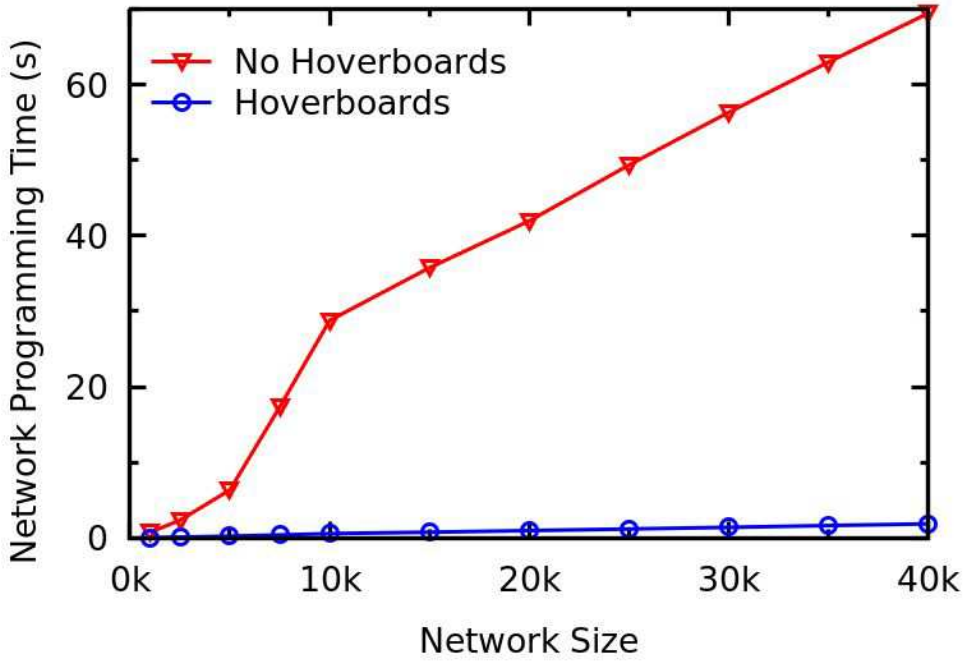
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Hoverboard Offloading



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Programming Time for Large Networks



Hoverboards reduce time to program network connectivity for large networks

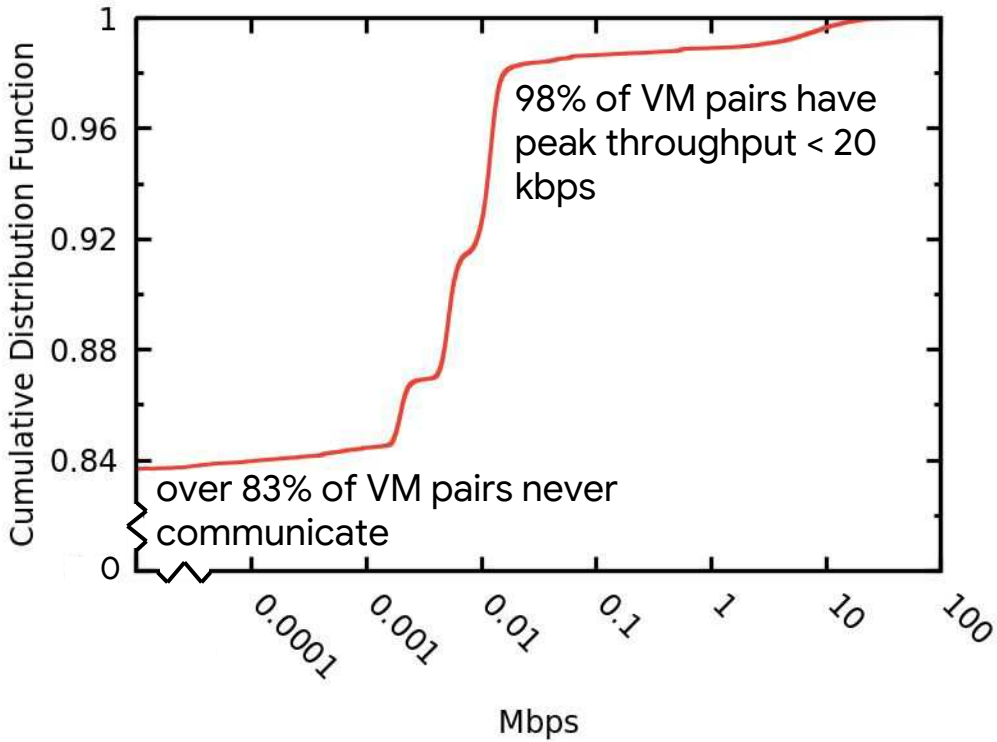
- 37X faster for a 40,000-VM network

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Why Hoverboards Are Effective

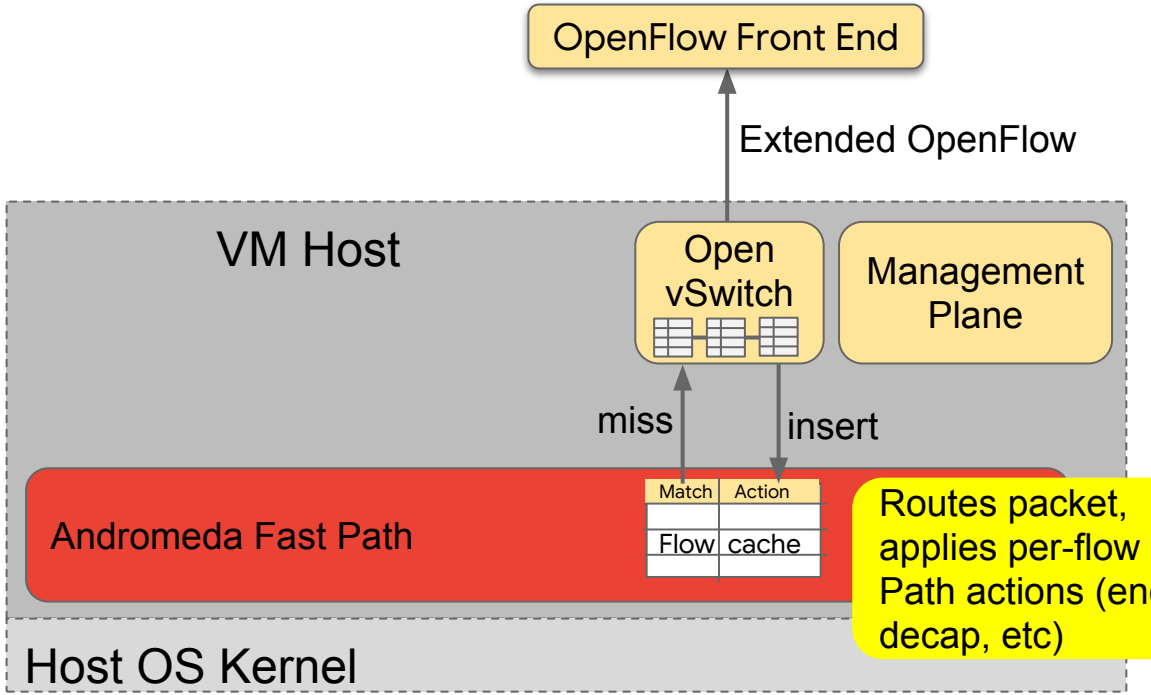
Peak throughput for all VM pairs in all virtual networks in one cluster over a 30-minute interval

Today, **more than 99.5% of traffic is offloaded**.



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Andromeda Data Plane

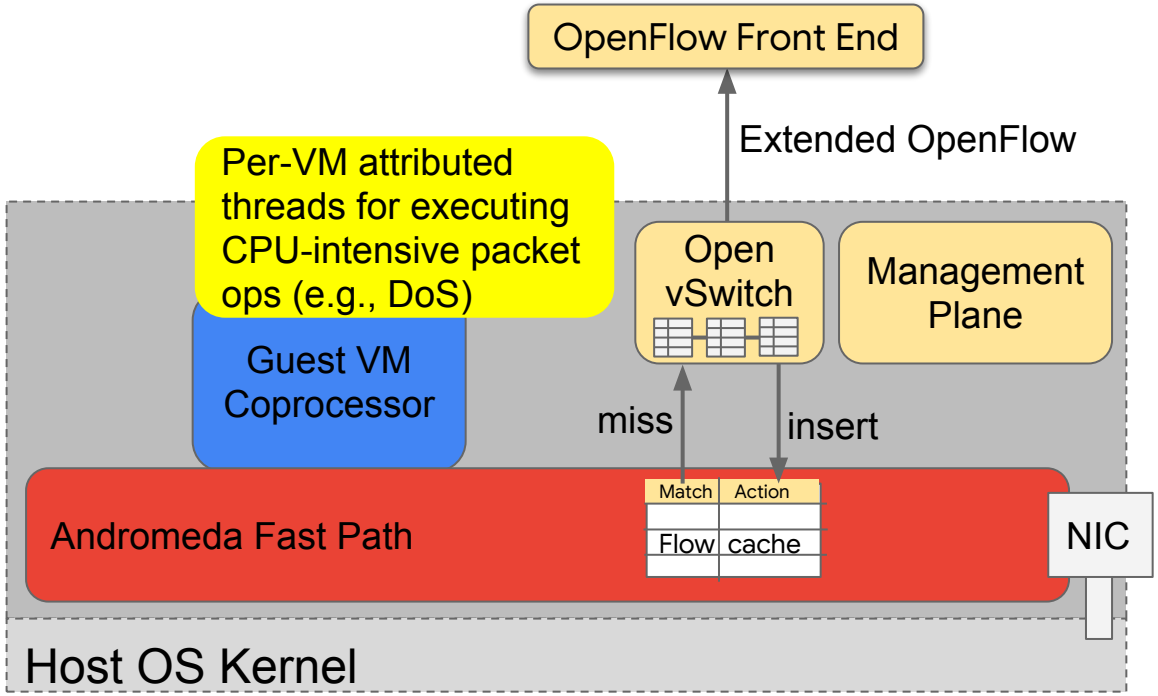


OS bypass, busy polling dedicated CPU Fast Path for **high performance**

Userspace dataplane, live migration, and bitless upgrades for **feature velocity**

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Andromeda Data Plane

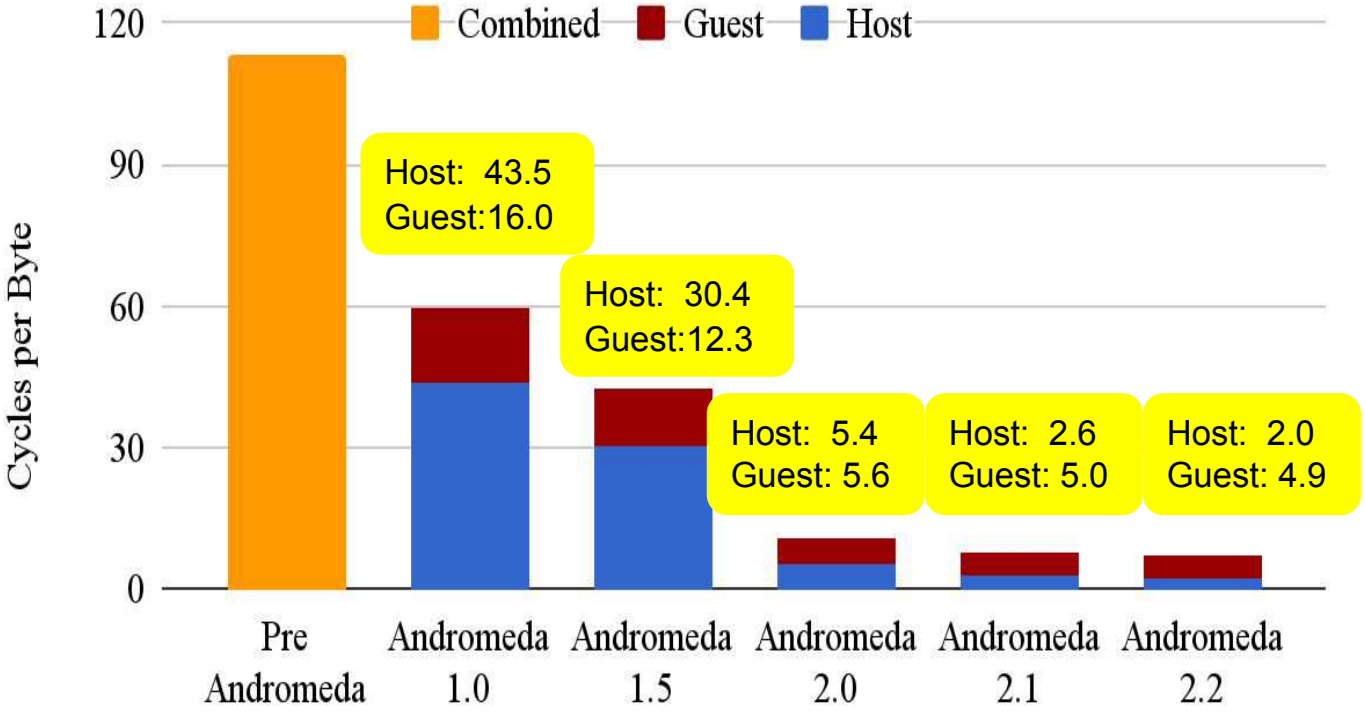


OS bypass, busy polling dedicated CPU Fast Path for **high performance**

Userspace dataplane, live migration, and hitless upgrades for **feature velocity**

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CPU Efficiency Evolution



Andromeda 1.0
Kernel datapath

Andromeda 1.5
Optimize pipeline

Andromeda 2.0
OS bypass, 1 thread hop

Andromeda 2.1
Remove thread hop

Andromeda 2.2
Memory copy offload

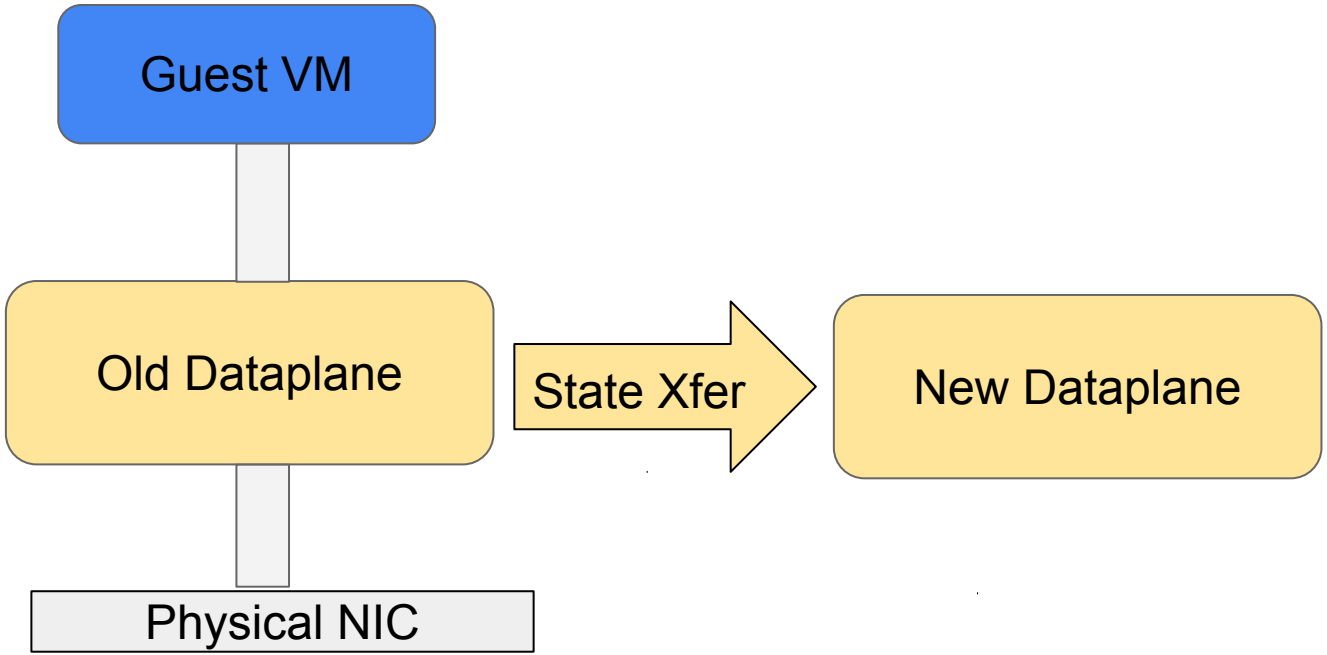
Velocity

A rapid release cycle enables swift deployment of features & bug fixes.

Our dataplane has **weekly** rollouts via non-disruptive upgrades.

Live migration allows VMs to be migrated between physical host without disruption, enabling transparent host maintenance.

Dataplane Hitless Upgrade 1 / 3



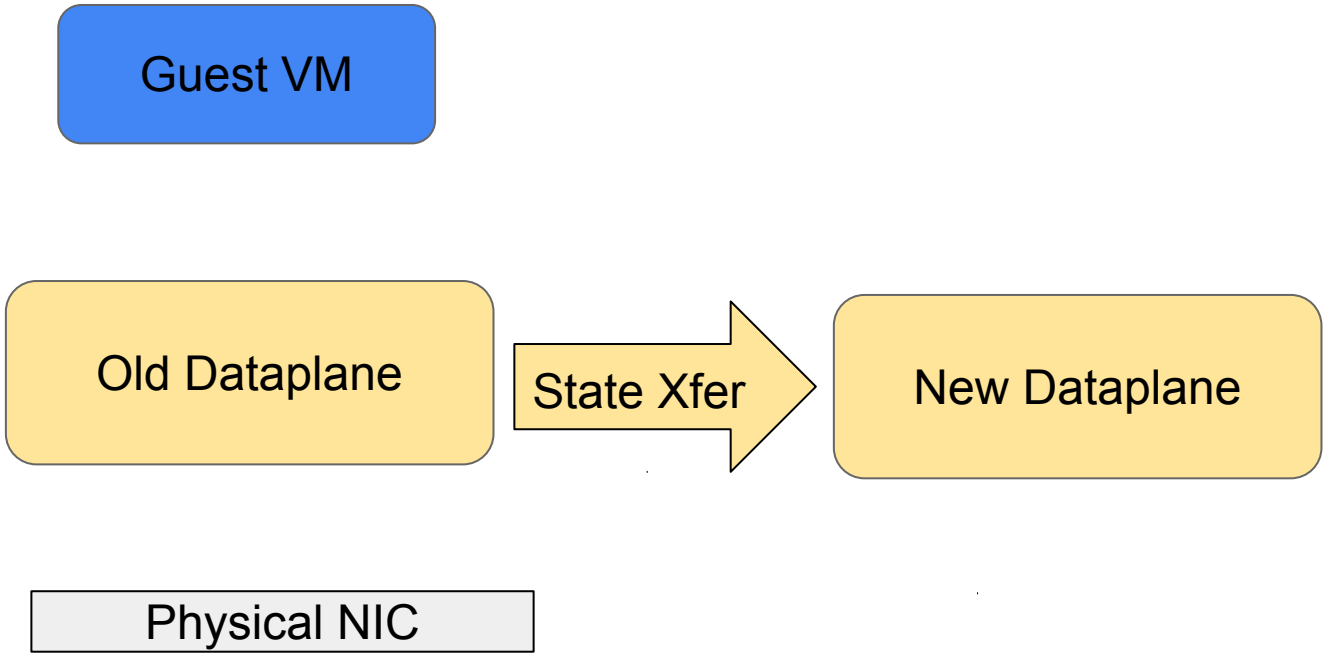
Upgrade Brownout

Old Dataplane state is transferred to New Dataplane in the background

Old Dataplane continues serving physical NIC & virtual NIC queues

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Dataplane Hitless Upgrade 2 / 3



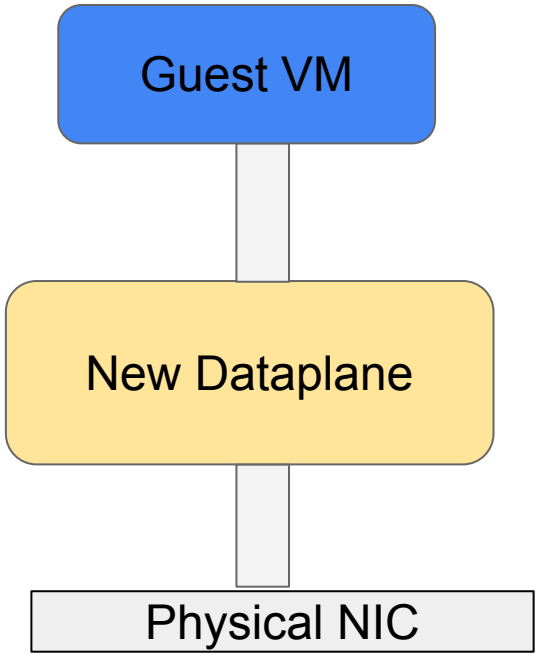
Upgrade Blackout

Old Dataplane stops serving virtual & physical NIC queues

Then, any updated (delta) Old Dataplane state is transferred to New Dataplane

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Dataplane Hitless Upgrade 3 / 3



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Upgrade Complete

State xfer done. Median blackout time is **270ms**.

New Dataplane starts serving VM virtual NIC & physical NIC queues

Old dataplane terminated

Conclusion

We have discussed the design and evolution of Andromeda

Control plane scalability & Rapid provisioning

- Hoverboard model avoids programming long tail of mostly idle flows on VM host. Scales to 100k VMs/network

High performance & Feature velocity

- OS Bypass dedicated CPU dataplane provides high performance (**> 30Gb/s**, **> 3M pps** with 1 core) & **weekly** non-disruptive updates